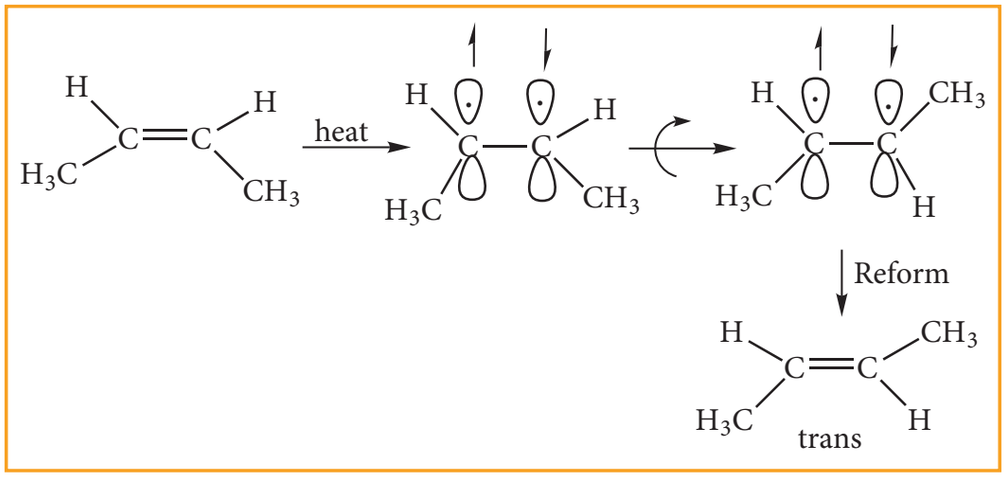
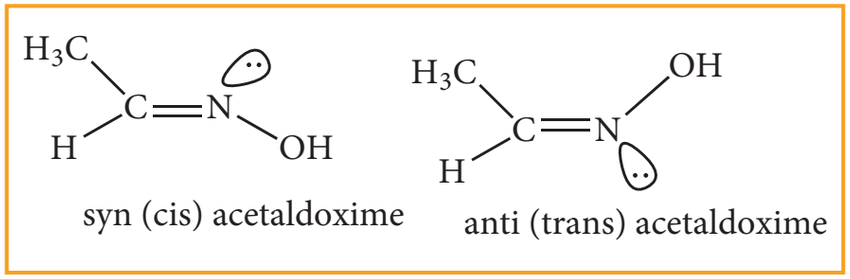



Generally the trans isomer is more stable than the corresponding *cis* isomers. This is because in the cis isomer, the bulky groups are on the same side of the double bond. The steric repulsion of the groups makes the cis isomers less stable than the trans isomers in which bulky groups are on the opposite side. These cis and trans isomers have different chemical property is. They can be separated by fractional distillation, gas chromatography etc., All alkenes with identical substrate do not show geometrical isomerism. Geometrical isomerism is possible only when each double bonded C atom is attached to two different atoms or groups eg. In propene no geometrical isomers are possible because one of the double bonded carbon has two identical H atoms.

Cis-trans isomerism is also seen around single bond. For eg: 1,3-butadiene has two double bonds in conjugation. $CH_2$=CH-CH=$CH_2$. It can exist in infinite number of conformations, but the following two extreme conformations are important.

## ii) Oximes and azo compounds:

Restricted rotation around C=N (oximes) gives rise to geometrical isomerism in oximes. Here 'syn' and 'anti' are used instead of cis and trans respectively. In the syn isomer the H atom of a doubly bonded carbon and –OH group of doubly bonded nitrogen lie on the same side of the double bond, while in the anti isomer, they lie on the opposite side of the double bond. For eg:






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